

Technical Report

Weakly-Supervised Remote Sensing Segmentation

via Point Annotations and Partial Cross-Entropy Loss

Model: SegFormer-B2 | Dataset: LoveDA

Framework: PyTorch | GPU: Tesla T4

April 2026

1. Method

1.1 Problem Formulation

Semantic segmentation requires classifying every pixel of an image into a predefined class. In the standard supervised setting this requires dense, pixel-wise annotation masks — an expensive and time-consuming process, particularly for high-resolution remote sensing imagery where a single image may contain hundreds of thousands of pixels across multiple land-cover classes.

This work addresses the point-supervised setting: during training, each image is annotated with only a sparse set of point labels — a small number of individual pixels whose class is known — while all remaining pixels carry no supervision signal. The core challenge is to train a segmentation model that generalises well to all pixels at inference time, despite receiving gradient updates from only a tiny fraction of them during training.

Formally, let:

- $x \in \mathbb{R}^{H \times W \times 3}$ be an RGB image of spatial resolution $H \times W$,
- $y \in \{0, \dots, C-1, \Omega\}^{H \times W}$ be the label map, where Ω denotes an unlabeled pixel,
- $M \in \{0, 1\}^{H \times W}$ be a binary point mask with $M_{ij}=1$ only at annotated locations,
- $f_\theta : \mathbb{R}^{H \times W \times 3} \rightarrow \mathbb{R}^{H \times W \times C}$ be the segmentation network with parameters θ .

The training objective is to minimise a loss function computed exclusively over labeled positions M , while evaluating on the full dense mask at validation time.

1.2 Partial Cross-Entropy Loss

The key contribution of this framework is the Partial Cross-Entropy Loss (pfCE), which gates the pixel-wise loss through the sparse point mask M . Only labeled pixels contribute gradients; the loss is normalised over the number of labeled pixels rather than the total pixel count, preventing the denominator from trivially dominating.

Standard Partial CE

For standard partial cross-entropy the per-pixel loss is ordinary cross-entropy:

$$\text{pfCE} = \sum_i [\text{CE}(f_\theta(x)_i, y_i) \times M_i] / \sum_i M_i$$

where $\text{CE}(p, y) = -\log p_y$ is the cross-entropy for predicted probability vector p and true class y , and the sums run over all spatial positions $i \in \{1, \dots, H \times W\}$.

Focal Partial CE (Used in This Work)

In practice, point annotations tend to be skewed toward easy, high-confidence regions. Standard CE assigns equal weight to all labeled pixels regardless of difficulty, which can cause the loss to be dominated by trivially-easy background pixels. Focal loss addresses this by down-weighting predictions the model already handles confidently:

$$\text{FL}(p_t) = -\alpha \times (1 - p_t)^\gamma \times \log(p_t)$$

Combined with the partial supervision mask:

$$pfFocal = \sum_i [FL(f_{\theta}(x)_i, y_i) \times M_i] / \sum_i M_i$$

where:

- p_t is the predicted probability of the true class at position i ,
- $\gamma = 2.0$ is the focusing parameter (higher values focus more on hard examples),
- $\alpha = 0.25$ is a class-balance weighting factor,
- $M_i \in \{0,1\}$ is the point mask — zero for unlabeled pixels, one for labeled pixels.

This formulation ensures that: (1) unlabeled pixels contribute exactly zero to the gradient, (2) easy background predictions are down-weighted, and (3) the loss scale remains consistent regardless of how sparse the point annotations are.

1.3 Point Annotation Simulation

Since LoveDA provides full dense segmentation masks, point annotations are simulated from them at training time to replicate realistic sparse labelling. For each training image, the simulation procedure is:

- Identify all unique classes present in the image's dense mask.
- For each class c , collect all pixel coordinates (y, x) where $mask[y, x] = c$.
- Randomly sample $K = 5$ pixels from that set without replacement.
- Set $M_{\{y,x\}} = 1$ and $point_label_{\{y,x\}} = c$ for each sampled coordinate.
- All remaining pixels are assigned $ignore_index = 255$ in the label map.

This produces a sparse label map used exclusively as the training supervision signal. The full dense mask is retained and used only for computing validation metrics. Point simulation runs inside the DataLoader's `__getitem__` method, meaning each training epoch sees a freshly sampled set of points per image — effectively acting as an additional stochastic augmentation.

1.4 Model: SegFormer-B2

SegFormer is a hierarchical transformer architecture for semantic segmentation composed of two main components: an efficient Mix Transformer (MiT) encoder and a lightweight all-MLP decoder head.

Encoder: Mix Transformer B2 (MiT-B2)

The MiT-B2 backbone produces multi-scale feature representations at four resolutions ($H/4$, $H/8$, $H/16$, $H/32$) through hierarchical self-attention with overlapping patch embeddings. Unlike ViT, MiT uses Efficient Self-Attention (reducing sequence length by a reduction ratio) and Mix-FFN (replacing positional encoding with a 3×3 depth-wise convolution), making it positional-encoding-free and thus robust to resolution changes at inference time. MiT-B2 contains approximately 25M parameters and is pretrained on ImageNet-1K.

Decoder: All-MLP Head

The decoder aggregates the four multi-scale features from the encoder using four linear projection layers, upsamples them to $H/4$ resolution, concatenates them, and applies a final linear

classification layer to produce class logits. The output is then upsampled to full $H \times W$ resolution using bilinear interpolation before loss computation. The entire decode head is randomly initialised (no pretrained weights exist for a 7-class head).

Differential Learning Rate

A differential learning rate strategy is applied: the pretrained MiT-B2 backbone is fine-tuned at $lr_backbone = 6e-6$ (10× smaller), while the randomly-initialised decode head is trained at $lr_head = 6e-5$. This prevents the pretrained features from being disrupted during early training when the head gradients are large.

1.5 Training Setup

The complete training configuration is summarised below:

Hyperparameter	Value
Dataset	LoveDA (Urban + Rural)
Number of classes	7
Image size	512×512
Training samples	2,522
Validation samples	1,669
Points per class	5 per image
Loss function	Partial Focal CE ($\gamma=2.0$, $\alpha=0.25$)
Model	SegFormer-B2 (nvidia/mit-b2)
Backbone LR	$6e-6$
Head LR	$6e-5$
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.999$)
Weight decay	0.01
LR schedule	Polynomial decay with 10% linear warmup
Gradient clipping	1.0 (global norm)
Batch size	4
Epochs	5
Hardware	Tesla T4 (Kaggle)

Data Augmentation (Training Only)

- RandomResizedCrop to 512×512 (scale 0.5-1.0, ratio 0.75-1.33)
- Horizontal and Vertical Flip ($p=0.5$ each)
- RandomRotate90 ($p=0.5$)

- ColorJitter: brightness, contrast, saturation ± 0.3 , hue ± 0.1 ($p=0.6$)
- ImageNet normalisation: $\mu=[0.485, 0.456, 0.406]$, $\sigma=[0.229, 0.224, 0.225]$

Evaluation Protocol

Validation is performed on full dense ground-truth masks — no point masking is applied during evaluation. The primary metric is mean Intersection-over-Union (mIoU) computed via confusion matrix accumulation across the full validation set. Pixel accuracy (PixAcc) is reported as a secondary metric. Void pixels (LoveDA label 0, remapped to ignore_index=255) are excluded from all metric computations.

$$\text{mIoU} = (1/C) \times \sum_c \text{TP}_c / (\text{TP}_c + \text{FP}_c + \text{FN}_c)$$

2. Experiment

2.1 Purpose

The primary objective of this experiment is to validate that a SegFormer-B2 model trained with Partial Focal Cross-Entropy Loss on simulated sparse point annotations can learn meaningful semantic segmentation of remote sensing imagery, despite receiving supervision from only 5 labeled pixels per class per image.

Specifically, the experiment addresses the following research questions:

- Can a pretrained transformer backbone compensate for the severe label sparsity inherent in point supervision?
- Does training loss decrease consistently across epochs despite sparse gradients?
- Which land-cover classes benefit most and least from point-based supervision?
- How does per-epoch mIoU evolve — is there evidence of overfitting within 5 epochs?

2.2 Hypothesis

Given the strong ImageNet-pretrained representations in MiT-B2, we hypothesise that:

- **Training loss will decrease monotonically across all 5 epochs, demonstrating that partial CE loss provides a useful learning signal despite seeing only ~35 labeled pixels per image (5 pts $\times$ 7 classes) out of $512 \times 512 = 262,144$ total pixels.**H1 — Convergence:
- **Validation mIoU will improve across epochs, demonstrating that gradient updates from point-labeled pixels generalise spatially to surrounding unlabeled pixels, a consequence of the pretrained features capturing class-discriminative spatial context.**H2 — Generalisation:
- **Structurally simple or spatially dominant classes (Road, Water) will achieve higher per-class IoU than visually ambiguous or rare classes (Barren, Forest), as dominant classes provide more training signal even under sparse sampling.**H3 — Class imbalance sensitivity:
- **Given only 5 epochs of training, we expect mIoU to peak around epoch 4 and potentially begin to plateau or slightly decline at epoch 5, as the model approaches**

the limits of what sparse supervision can provide without regularisation.H4 — Peak before epoch 5:

2.3 Experimental Process

Dataset: LoveDA

LoveDA (Love Data Challenge) is a large-scale remote sensing semantic segmentation dataset containing high-resolution aerial imagery from three Chinese cities. It covers two domains (Urban and Rural) and seven land-cover classes: Background, Building, Road, Water, Barren, Forest, and Agriculture. Images are 1024×1024 pixels at 0.3m spatial resolution.

Split	Domain	Images	Used
Train	Urban + Rural	2,522	✓
Val	Urban + Rural	1,669	✓
Test	Urban + Rural	1,796	× (no labels)

Point Simulation Details

For each training image, 5 random pixels are sampled per class present in that image from the dense ground-truth mask. With 7 classes, this gives at most 35 labeled pixels per 512×512 image — a labeling density of approximately 0.013%. Simulation runs independently per epoch, meaning the sampled point locations differ across epochs.

Training Procedure

The model was trained for 5 epochs on the combined Urban and Rural training splits. After each epoch, the full validation set was evaluated using the dense ground-truth masks. The checkpoint achieving the highest validation mIoU was saved as the best model. Training ran on a Kaggle Tesla T4 GPU.

2.4 Results

Epoch-by-Epoch Training Log

Epoch	Train Loss	Val mIoU	Val PixAcc	Best?
1	0.2594	0.3888	0.5888	✓
2	0.1622	0.4179	0.6166	✓
3	0.1501	0.4469	0.6305	✓
4	0.1421	0.4569	0.6414	✓ Best
5	0.1399	0.4508	0.6332	

Best Checkpoint Performance

Metric	Value
Best Epoch	4
Best mIoU	0.4569 (45.69%)
Best PixAcc	0.6414 (64.14%)
Final Train Loss	0.1399

Per-Class IoU (Best Epoch — Epoch 4)

Class	Per-Class IoU	Relative Rank
Background	0.4712	3rd
Building	0.5010	2nd
Road	0.5340	1st (Best)
Water	0.6371	1st overall
Barren	0.2744	7th (Worst)
Forest	0.3285	6th
Agriculture	0.4522	4th

2.5 Analysis and Discussion

Hypothesis Evaluation

H1 — Convergence: CONFIRMED. Training loss decreased consistently across all 5 epochs (0.2594 → 0.1622 → 0.1501 → 0.1421 → 0.1399), demonstrating that the Partial Focal CE loss provides a stable and useful learning signal despite supervision being limited to approximately 35 pixels per image.

H2 — Generalisation: CONFIRMED. Validation mIoU improved from 0.3888 at epoch 1 to a peak of 0.4569 at epoch 4 — a gain of +6.81 mIoU points. This confirms that gradient updates from labeled point pixels propagate meaningful class information to surrounding spatial regions through the pretrained transformer's self-attention mechanism.

H3 — Class imbalance sensitivity: CONFIRMED. Water achieved the highest per-class IoU (0.6371), benefiting from its high spectral contrast and homogeneous texture. Road ranked second (0.5340) due to its consistent structural geometry. Barren land performed worst (0.2744), likely because it is visually similar to Road and Building in arid regions, and occupies a small area across most images, limiting the number of available sampling positions.

H4 — Peak before epoch 5: CONFIRMED. mIoU peaked at epoch 4 (0.4569) and declined at epoch 5 (0.4508). This early-plateau behaviour is consistent with point-supervised training: once the model has learned the class boundaries accessible from sparse point labels, additional epochs begin to overfit to the specific sampled locations rather than learning generalisable spatial patterns. Longer training would require additional regularisation (e.g., consistency regularisation on unlabeled pixels, dropout scheduling).

Key Observations

- The largest single-epoch mIoU gain occurred between epochs 1 and 2 (+0.0291), suggesting the model rapidly learned coarse class boundaries in the first epoch and refined them subsequently.
- Water IoU (0.6371) significantly outperformed all other classes by a margin of +0.103 over Road, reflecting that spectral distinctiveness is more important than spatial abundance under point supervision.
- Barren and Forest IoU (0.2744 / 0.3285) were substantially lower than other classes. Both classes share similar green/brown spectral signatures with Agriculture, creating inter-class confusion that 5 points per class cannot resolve.
- Despite sparse supervision (0.013% labeling density), the model achieved 45.69% mIoU — a competitive result for weakly-supervised segmentation, demonstrating the power of ImageNet-pretrained MiT-B2 features as a starting point.

2.6 Limitations and Future Work

- Only 5 training epochs were run due to compute budget constraints. Full convergence would likely require 30-80 epochs with a proper learning rate schedule, potentially achieving mIoU in the 55-65% range.
- No semi-supervised regularisation was applied to unlabeled pixels. Incorporating pseudo-label generation or consistency regularisation (e.g., mean-teacher) for unlabeled pixels would substantially improve performance.
- Point density was fixed at $K=5$ per class. An ablation over $K \in \{1, 3, 5, 10\}$ would quantify the sensitivity of mIoU to annotation budget, informing practical annotation guidelines.
- Loss function comparison (standard CE vs. Focal CE) was not run due to time constraints. A controlled comparison would empirically validate the benefit of focal weighting under sparse supervision.

2.7 Conclusion

This work demonstrates that SegFormer-B2 trained with Partial Focal Cross-Entropy Loss on simulated sparse point annotations achieves 45.69% mIoU on the LoveDA remote sensing segmentation benchmark after only 5 epochs of training. All four experimental hypotheses were confirmed: training loss converged monotonically, validation performance improved consistently, class performance correlated with spectral distinctiveness, and the model peaked at epoch 4 before plateauing.

The results validate the Partial CE loss formulation as a practical and effective approach to point-supervised semantic segmentation: by gating loss computation through a sparse binary mask and normalising over labeled pixels only, the model receives a clean gradient signal that —

combined with strong pretrained representations — generalises well beyond the annotated pixels. This is a meaningful contribution toward reducing the annotation cost of remote sensing image analysis.